A Project Report

On

Explainable AI for Clinical Decision Support in Heart Disease Prediction

Submitted in partial fulfillment of the

requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

GALGOTIAS UNIVERSITY

(Established under Galgotias University Uttar Pradesh Act No. 14 of 2011)

DEGREE

Session 2025-26
in

Computer Science and Engineering

By
Avi Jain
(22SCSE1410010)
Durgesh Kumar Shukla
(22SCSE1410002)

Dr. Santosh Kumar Sir

SCHOOL OF COMPUTING SCIENCE AND ENGINEERING DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

GALGOTIAS UNIVERSITY, GREATER NOIDA
INDIA

April, 2026

SCHOOL OF COMPUTING SCIENCE AND ENGINEERING
GALGOTIAS UNIVERSITY, GREATER NOIDA

CANDIDATE'S DECLARATION

I/We hereby certify that the work which is being presented in the project, entitled **"Explainable AI for Clinical Decision Support in Heart Disease Prediction"** in partial fulfillment of the requirements for the award of the B. Tech. (Computer Science and Engineering) submitted in the School of Computing Science and Engineering of Galgotias University, Greater Noida, is an original work carried out during the period of Aug, 2025 to Jun 2026, under the supervision of Prof **Dr. Santosh Kumar** Department of Computer Science and Engineering, of School of Computing Science and Engineering , Galgotias University, Greater Noida.

The matter presented in the thesis/project/dissertation has not been submitted by me/us for the award of any other degree of this or any other places.

Avi Jain (22SCSE1410010)

Durgesh Kumar Shukla (22SCSE1410002)

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Dr. Santosh Kumar

Professor

CERTIFICATE

This is to certify that Project Report entitled "**Explainable AI for Clinical Decision Support in Heart Disease Prediction**" which is submitted by Avi Jain, Durgesh Kumar Shukla in partial fulfillment of the requirement for the award of degree B.Tech. in Department of Computer Science and Engineering, Galgotias University, Greater Noida, India is a record of the candidate own work carried out by him/them under my supervision. The matter embodied in this thesis is original and has not been submitted for the award of any other degree

Signature of Examiner(s) **Signature of Supervisor(s)**

External Examiner **Signature of Program Chair**

Date:

Place: Greater Noida

ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the Project undertaken during B.Tech Final Year. We owe special debt of gratitude to Professor **Dr. Santosh Kumar** *Department of Computer Science & Engineering,* Galgotias University, Greater Noida, India *for his constant support and guidance throughout the course of our work. His/Her sincerity, thoroughness and perseverance have been a constant source of inspiration for us. It is only his consistent efforts that our endeavors have seen light of the day.*

We also take the opportunity to acknowledge the contribution of Professor (Dr.) Harish Kumar G R, Dean, Department of Computer Science & Engineering, Galgotias University, Greater Noida, India *for his full support and assistance during the development of the project.*

We also do not like to miss the opportunity to acknowledge the contribution of all faculty members of the department for their kind assistance and cooperation during the development of our project. Last but not the least, we acknowledge our friends for their contribution in the completion of the project.

Signature:

Name: Avi Jain

Roll No.: 22SCSE1410010

Name: Durgesh Kumar Shukla

Roll No.: 22SCSE1410002

Date:

ABSTRACT

Cardiovascular diseases are among the leading causes of death globally, emphasizing the need for timely and accurate diagnosis. In the medical field, Artificial Intelligence (AI) has shown immense potential to assist healthcare professionals in predicting and preventing such conditions. However, most AI-based systems operate as “black boxes,” providing predictions without explaining the underlying reasoning. This lack of transparency often limits their acceptance in clinical settings.

This project, titled **“Explainable AI for Clinical Decision Support in Heart Disease Prediction,”** aims to bridge this gap by combining predictive modeling with interpretability. The system utilizes a **Random Forest classifier** trained on the **UCI Cleveland Heart Disease dataset** to predict the likelihood of a patient having heart disease based on clinical parameters such as age, cholesterol level, resting blood pressure, and chest pain type.

To ensure transparency, the project integrates **SHAP (SHapley Additive exPlanations)**, an explainability technique that visualizes how each feature influences the model’s decision. The user-friendly **Flask web interface** allows clinicians or users to input patient data, receive predictions (High/Low risk), and view SHAP plots showing feature contributions

TABLE OF CONTENTS Page

CHAPTER-1

INTRODUCTION

1.1 Problem Introduction

Heart disease remains one of the leading causes of mortality worldwide. Traditional diagnostic methods often rely on manual interpretation of various health indicators, which may be time-consuming and prone to human error. **Artificial Intelligence (AI)** can assist by identifying complex patterns in patient data to predict disease risk. However, black-box AI models often lack transparency, making them difficult to trust in medical contexts. Hence, there is a pressing need **for Explainable AI (XAI) systems** that can not only make accurate predictions but also justify them in understandable terms for clinicians.

1.1.2 Project Objective

The main objective of this project is to develop a prototype that:

- Predicts the likelihood of heart disease using patient data.
- Provides interpretable explanations of model predictions using SHAP.
- Offers an easy-to-use web interface for clinical decision support.
- Demonstrates the integration of explainable AI techniques into healthcare workflows.

1.1.3 Scope of the Objective

The project focuses on developing a proof-of-concept prototype for heart disease prediction. It uses the UCI Cleveland Heart Disease dataset for model training and testing. The web application, built with Flask, allows users to input parameters and visualize both predictions and their SHAP-based explanations. The system is designed to be extendable to real-world healthcare data and can support further model and UI enhancements.

CHAPTER 2

REQUIREMENTS SPECIFICATION

2.1 Product Perspective

The proposed system is a standalone, research-level web application intended to serve as a decision-support tool for healthcare professionals, specifically cardiologists and general practitioners. It is not a diagnostic tool; rather, it provides supplementary, evidence-based information to aid the clinician's decision-making process. The system operates "offline" by using a pre-trained ML model and does not require a live internet connection for its core logic, though it is delivered via a web interface.

2.1.1 System Interfaces

User Interface (UI): The primary interface is a Graphical User Interface (GUI) rendered in a standard web browser.

- Input Page: A secure web form where the clinician can manually enter a patient's data (e.g., age, sex, cholesterol, resting blood pressure, etc.).

- Output/Results Page: A dashboard that appears after submitting the data, displaying the prediction and explanation.

2.1.2 Interfaces

(This section is often interpreted as external data or hardware interfaces. Given 2.1.1 and 2.1.3, this section will define the data and model interfaces.)

- Data Interface: The system interfaces with a static, pre-processed dataset (e.g., data.csv) during the training phase. During the inference phase (live prediction), it only interfaces with the data provided by the user via the web form.

- Model Interface: The web application's backend interfaces with the saved, serialized ML model (e.g., a model.pkl file) and the corresponding XAI explainer.

2.1.3 Software Interfaces

- **Frontend:** HTML templates rendered using Flask’s Jinja2.
- **Backend:** Flask API integrating Python scripts and the trained model.
- **Visualization:** Matplotlib used for SHAP value plots.

2.2 Product Functions

- Accepts user input of 13 medical parameters.
- Predicts heart disease risk (Low/High).
- Displays prediction probability.
- Generates SHAP bar plot showing feature impact on prediction.

2.3 User Characteristics

Intended users include medical practitioners, researchers, and students exploring clinical AI applications. Users are expected to have basic knowledge of health parameters.

2.4 Constraints

- Model trained on a limited dataset (UCI Cleveland).
- Not certified for real medical use.
- Requires numerical input formatting.
- SHAP visualization may take extra computation time.

2.5 Assumptions and Dependencies

- Dataset is accurate and representative.
- Users input valid numeric values.
- Dependencies include Flask, scikit-learn, SHAP, matplotlib, and joblib.

CHAPTER 3

SYSTEM DESIGN

3.1 System Architecture Overview

The architecture of the project "Explainable AI for Clinical Decision Support in Heart Disease Prediction" is designed in a modular three-tier structure, comprising the Data Layer, Application Logic Layer, and Presentation Layer. Each layer performs a distinct function, ensuring modularity, scalability, and maintainability of the system.

At its core, the system integrates a machine learning model for disease prediction with an explainability module (SHAP) for model interpretation. The backend communicates with the trained model and explainability framework, while the frontend interface enables end-users (clinicians or students) to interact with the system through a web browser.

3.2 Architectural Layers

1. Data Layer

This layer handles all data-related operations including dataset acquisition, preprocessing, and model training.

- The system uses the UCI Cleveland Heart Disease Dataset, containing medical features such as age, sex, chest pain type, resting BP, cholesterol, and more.
- Missing values (denoted by '?') are handled using pandas for data cleaning.
- The dataset is split into training and testing subsets (80–20 ratio).
- A Random Forest Classifier is trained to learn patterns and relationships between clinical features and heart disease occurrence.
- The trained model and metadata (feature names) are stored in a serialized file model.joblib using joblib, allowing the Flask app to load it efficiently during inference.

Key Components:

- trainmodel.py script (handles model training)
- heart.csv dataset file
- model.joblib (stored model for deployment)

---

2. Application Logic Layer

This layer acts as the core processing engine of the application. It is implemented using Flask, a lightweight Python web framework.

- The Flask app (app.py) manages HTTP requests, data validation, and interaction with the trained model.
- When the user submits patient information via the web form, the app performs the following steps:
    1. Input Processing: Retrieves 13 numerical input values corresponding to medical features.
    2. Prediction: Loads the trained model and computes the probability of heart disease using model.predict_proba().
    3. Explainability: Invokes SHAP's TreeExplainer to compute Shapley values for the given input, quantifying each feature's contribution to the prediction.
    4. Visualization: Uses matplotlib to generate a SHAP bar chart highlighting the most influential features in descending order of importance.
    5. Response Rendering: The computed prediction, probability, and SHAP image are returned to the frontend for display.

This layer essentially bridges the AI model and the web interface, ensuring smooth data flow and interpretability output generation.

Key Components:

- Flask routes (/ endpoint)
- SHAP TreeExplainer integration
- Matplotlib visualizations
- Dynamic rendering with Jinja2 templates

---

3. Presentation Layer

This layer serves as the user interface, allowing users to interact with the AI model intuitively.

- The HTML frontend (template index.html) displays an input form with labeled fields for all 13 medical attributes.
- Upon submission, the page displays:
    - Prediction result: "High Risk" or "Low Risk"

    - Probability score: Likelihood of heart disease (e.g., 0.784)
    - SHAP Bar Chart: Explains which features contributed positively or negatively to the model's decision.
- The interface is designed for simplicity and usability, suitable for quick clinical assessments or educational demonstrations.

Key Components

- HTML + Jinja2 templates for rendering results
- Static folder for SHAP images (static/shap.png)
- CSS (optional styling for readability)

CHAPTER 4

IMPLEMENTATION

4.1 Introduction

This chapter details the system implementation, including model training, backend logic, and web interface integration.

4.2 System Implementation Overview

- The model is trained using trainmodel.py with a Random Forest Classifier.
- After training, the model is serialized as model.joblib.
- The Flask app (app.py) loads the model, takes user input, computes predictions, and generates SHAP visualizations.

4.3 Tools and Technologies Used

Category	Tools/Frameworks
Programming Language	Python
Framework	Flask
Libraries	scikit-learn, SHAP, matplotlib, pandas, numpy
Model	Random Forest Classifier
Dataset	UCI Cleveland Heart Disease Dataset
Deployment	Localhost Flask Server

4.4 Implementation Steps

1. Download and preprocess dataset.
2. Train Random Forest model using scikit-learn.
3. Save model using joblib.
4. Build Flask web interface for user input.
5. Integrate SHAP explainability and visualize feature impact.
6. Test prototype with sample data.

4.5 Result and Performance Analysis

The Random Forest classifier achieved an accuracy of approximately **85–88%** on test data. SHAP visualizations successfully highlighted key predictors such as **chest pain type, cholesterol, resting BP**, and **maximum heart rate achieved**, allowing transparent interpretation of the model's reasoning.

CHAPTER 5

CONCLUSION

5.1 Conclusion and Future Scope

The developed system demonstrates the potential of **Explainable AI in clinical decision support**. By combining predictive modeling with interpretability, clinicians can make informed and transparent diagnostic decisions. Future enhancements could include:

- Integration with real-world hospital EHR data.
- Use of deep learning models with integrated explainability.
- Deployment on cloud platforms for scalability.
- Improved visualization dashboards for clinicians.

ABSTRACT

Heart disease continues to be the leading cause of mortality worldwide, making early and accurate detection a critical challenge for modern healthcare systems. Traditional machine learning approaches, while effective in predictive accuracy, suffer from a significant lack of transparency that hinders their adoption in clinical settings. This project presents a comprehensive Clinical Decision Support System (CDSS) designed to predict the risk of heart-related diseases while simultaneously providing clinically meaningful explanations for its predictions.

The proposed system integrates multiple machine learning models including Logistic Regression, Random Forest, Support Vector Machine (SVM), Gradient Boosting, and XGBoost, which are individually evaluated and then combined into a stacked ensemble architecture. The stacked ensemble model significantly outperforms all individual models, achieving an accuracy of 96.0% and precision of 95.9% on the CDC BRFSS 2020 dataset containing over 400,000 patient records.

To address the black-box nature of complex machine learning models, the system incorporates SHAP (SHapley Additive exPlanations), an Explainable Artificial Intelligence (XAI) technique based on game-theoretic principles. SHAP provides both local and global interpretability, revealing the contribution of each clinical feature — such as age, BMI, general health status, chest pain type, and physical activity — to individual predictions.

The system was evaluated on two large-scale datasets: the Diabetes 130-US Hospital dataset and the CDC BRFSS 2020 Annual Survey. Experimental results confirm that the stacked ensemble model achieves superior performance compared to all baseline models. The integration of SHAP-based visualizations enables doctors to validate model predictions against established medical knowledge, thereby increasing trust, transparency, and clinical utility.

This research demonstrates that combining high-accuracy ensemble learning with explainable AI techniques yields a practical, transparent, and trustworthy decision support framework for heart disease prediction. Future work may extend this to include temporal health records and larger multi-institutional datasets.

Keywords: Clinical Decision Support System, Explainable Artificial Intelligence (XAI), SHAP, Machine Learning, Heart Disease Prediction, Stacked Ensemble Model, Random Forest, XGBoost, Gradient Boosting

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
ML	Machine Learning
CDSS	Clinical Decision Support System
SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-agnostic Explanations
LR	Logistic Regression
RF	Random Forest
SVM	Support Vector Machine
GB	Gradient Boosting
XGB	Extreme Gradient Boosting
CDC	Centers for Disease Control and Prevention
BRFSS	Behavioral Risk Factor Surveillance System
EHR	Electronic Health Record
BMI	Body Mass Index
F1	F1-Score (Harmonic Mean of Precision and Recall)
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Heart disease remains the leading cause of mortality globally, claiming millions of lives annually and placing enormous strain on healthcare systems worldwide. According to the World Health Organization (WHO), cardiovascular diseases account for approximately 17.9 million deaths each year, representing 31% of all global deaths. The ability to identify high-risk patients early and intervene in a timely manner is therefore one of the most pressing challenges in modern medicine.

The rapid growth of healthcare data — encompassing electronic health records (EHRs), laboratory results, imaging data, and survey responses — has created an unprecedented opportunity for the application of artificial intelligence (AI) and machine learning (ML) in medical diagnosis and prognosis. These techniques can uncover complex, non-linear patterns in large datasets that would be impossible for human clinicians to detect manually.

However, despite their impressive predictive performance, most machine learning models operate as 'black boxes' — producing predictions without revealing the reasoning behind them. This opacity is a fundamental barrier to clinical adoption. Physicians cannot responsibly rely on a system that simply outputs a risk score without explaining which factors drove that prediction, how confident the model is, or how the result aligns with established medical knowledge.

Explainable Artificial Intelligence (XAI) has emerged as a solution to this fundamental challenge. XAI refers to a set of techniques and methodologies designed to make the decisions of AI models understandable and interpretable to human users. Among XAI methods, SHAP (SHapley Additive exPlanations) has gained significant traction in medical applications due to its strong theoretical foundation in cooperative game theory and its ability to provide both local (patient-specific) and global (population-level) explanations.

This project is motivated by the intersection of three critical needs: the clinical urgency of early heart disease detection, the availability of large-scale population health datasets, and the emerging capability of XAI to bridge the gap between algorithmic prediction and clinical interpretability.

1.2 Problem Statement

The central problem addressed in this research is twofold. First, while machine learning models demonstrate high accuracy in predicting heart disease from clinical and demographic data, their use in real-world clinical practice is severely limited by their lack of interpretability. Clinicians require not just a prediction, but a justification that aligns with their medical understanding.

Second, existing heart disease prediction systems typically focus on a single model or a narrow set of features, and seldom integrate XAI techniques in a systematic and user-friendly manner. There is a gap in the literature for a comprehensive, end-to-end framework that combines the predictive power of ensemble learning with the explanatory capability of SHAP to support clinical decision-making.

Formally, the problem can be stated as: Given a set of clinical and demographic features for a patient, develop a machine learning model that can accurately predict the binary outcome of heart disease presence or absence, while simultaneously providing transparent, feature-level explanations that are clinically meaningful and actionable for healthcare professionals.

1.3 Objectives of the Study

The primary objectives of this research project are as follows:

- To conduct a comprehensive review of existing machine learning and explainable AI approaches in heart disease prediction.
- To preprocess and analyze large-scale public health datasets, including the CDC BRFSS 2020 Survey and the Diabetes 130-US Hospital dataset.
- To implement and evaluate multiple machine learning classifiers: Logistic Regression, Random Forest, Support Vector Machine, Gradient Boosting, and XGBoost.
- To design and evaluate a stacked ensemble model that combines the strengths of multiple individual classifiers for improved predictive performance.
- To integrate SHAP-based explainable AI to provide transparent, feature-level explanations for individual predictions.
- To develop a graph-based visualization module that illustrates dependencies between clinical risk factors.
- To compare the performance of the proposed system against state-of-the-art individual models reported in the literature.

1.4 Scope of the Work

This project focuses on binary classification of heart disease using structured tabular healthcare data. The scope includes supervised learning using publicly available datasets, ensemble learning through stacking, and post-hoc explainability using SHAP. The system is designed to support clinicians and researchers in understanding risk factors at both the individual patient level and the population level.

The scope does not include real-time deployment in a hospital environment, integration with live EHR systems, or the use of imaging or genomic data. Future extensions may address these areas.

1.5 Significance of the Study

The significance of this research lies in its potential to bridge the persistent gap between algorithmic accuracy and clinical adoption in cardiovascular medicine. Despite the proliferation of high-accuracy machine learning models in academic research, the actual deployment of AI-based diagnostic tools in routine clinical workflows remains remarkably limited. A 2024 survey of US cardiologists found that fewer than 12% regularly use any AI-based decision support tool, with the most frequently cited barrier being a lack of trust in opaque algorithmic predictions.

By integrating explainable AI directly into the prediction pipeline, this work contributes to the broader movement toward responsible, human-centered AI in healthcare. The proposed framework demonstrates that interpretability does not need to come at the cost of predictive performance — a misconception that has historically discouraged the adoption of complex ensemble methods in clinical settings.

Furthermore, the study has practical significance for public health policy. By identifying the most influential risk factors at the population level, the SHAP analysis can inform targeted prevention campaigns, screening priorities, and resource allocation. For low- and middle-income countries where cardiovascular disease burden is rising rapidly but specialist access is limited, an AI-based screening tool with transparent reasoning could meaningfully extend the reach of preventive cardiology.

1.6 Heart Disease: Epidemiology and Risk Factors

To contextualize the technical contributions of this work, it is useful to briefly review the epidemiology of heart disease and its established risk factors. Heart disease, also known as cardiovascular disease (CVD), is an umbrella term that includes coronary artery disease (CAD), heart failure, arrhythmias, valvular heart disease, and congenital heart defects. Coronary artery disease — caused by atherosclerotic plaque buildup in the coronary arteries — is the most common form and accounts for the majority of cardiovascular deaths globally.

The risk factors for heart disease are conventionally classified into modifiable and non-modifiable categories. Non-modifiable risk factors include age, sex (males have higher risk at younger ages), race/ethnicity, and family history of premature CVD. Modifiable risk factors include hypertension, dyslipidemia, diabetes mellitus, obesity, tobacco use, physical inactivity, unhealthy diet, excessive alcohol consumption, and chronic psychosocial stress.

Table 1.1 below summarizes the major risk factors for heart disease and their typical relative risk (RR) impact, as established by large-scale epidemiological studies such as the Framingham Heart Study and the INTERHEART study. These risk factors form the conceptual basis for the feature set used in this research.

Risk Factor	Category	Approximate Relative Risk

Age (>65 years)	Non-modifiable	RR 4–8x compared to <45 years
Male Sex	Non-modifiable	RR 1.5–2x at younger ages
Hypertension	Modifiable	RR 2–3x for stroke and MI
Diabetes Mellitus	Modifiable	RR 2–4x for CVD events
Tobacco Smoking	Modifiable	RR 2–3x for CAD
High LDL Cholesterol	Modifiable	RR 1.5–2x per 1 mmol/L
Obesity (BMI > 30)	Modifiable	RR 1.5–2x for CHD
Physical Inactivity	Modifiable	RR 1.3–1.5x
Family History of Early CVD	Non-modifiable	RR 1.5–2x
Chronic Kidney Disease	Modifiable*	RR 2–3x

Table 1.1: Major Risk Factors for Heart Disease (Source: Framingham/INTERHEART)

These epidemiological insights guide the interpretation of the SHAP-based feature importance results presented in Chapter 5. A key validation criterion for any XAI system in cardiovascular medicine is that its top-ranked features should align broadly with the established clinical risk factors above.

1.7 Report Organization

The remainder of this report is organized as follows. Chapter 2 presents a detailed literature review of related work in machine learning for heart disease prediction and XAI techniques in healthcare. Chapter 3 describes the methodology, including dataset description, preprocessing, feature engineering, and the individual machine learning models. Chapter 4 presents the proposed framework, including the stacked ensemble architecture, SHAP integration, and visualization module. Chapter 5 documents the implementation details, including the development environment, software stack, hyperparameter tuning, and reproducibility considerations. Chapter 6 reports experimental results and comparative analysis, including statistical significance testing and error analysis. Chapter 7 concludes the report with a discussion of limitations and directions for future research.

CHAPTER 2

LITERATURE REVIEW

2.1 Traditional Machine Learning in Heart Disease Prediction

The application of machine learning to cardiovascular risk prediction has been an active research area for over two decades. Early studies primarily employed statistical models such as logistic regression and linear discriminant analysis, which offered interpretability but were limited in their ability to capture complex, nonlinear relationships in clinical data.

Asih et al. [10] proposed an interpretable machine learning model for heart disease prediction using logistic regression, demonstrating that classical models can achieve reasonable accuracy (88.2%) while maintaining transparency through coefficient analysis. However, their work also highlighted the inherent limitations of linear models when dealing with high-dimensional, nonlinear health data.

Random Forest classifiers have shown significant improvements over logistic regression in cardiovascular prediction tasks. Chen et al. [14] demonstrated that tuned Random Forest models with class balancing achieve accuracy levels around 91.2%, leveraging the ensemble nature of decision trees to reduce variance and improve generalization. The feature importance scores provided by Random Forest offer a basic level of interpretability, though they do not capture interaction effects or provide patient-level explanations.

Support Vector Machines (SVMs) have also been widely studied for heart disease classification. Liu et al. [15] applied SVM with RBF kernel and hyperparameter optimization (C and γ tuning), achieving accuracy of 91.8% on structured clinical datasets. SVMs are effective in high-dimensional spaces but are inherently black-box models, offering no direct insight into the decision process.

Gradient boosting methods represent the current state of the art in tabular data classification. Fitriyani et al. [17] proposed a novel approach utilizing bagging, histogram gradient boosting, and advanced feature selection for predicting cardiovascular diseases, achieving 90.7% accuracy. Sheridan et al. [16] demonstrated that XGBoost with advanced regularization and boosting rounds reaches 93.6% accuracy, making it one of the most powerful single-model approaches available.

Despite these advances, all of the above models share a common limitation: they do not inherently explain why a prediction was made. This is a critical gap when considering deployment in clinical environments where patient safety and regulatory compliance require justifiable decisions.

2.2 Explainable AI Techniques in Healthcare

The growing importance of transparency in AI-driven medical systems has spurred substantial research in explainable AI (XAI). XAI methods can broadly be categorized as intrinsically interpretable models (such as decision trees and logistic regression) and post-hoc explanation methods (such as SHAP, LIME, and feature permutation importance).

Abbas et al. [1] conducted a comprehensive review of XAI methods in clinical decision support systems, covering 2025 literature. Their work surveyed SHAP, LIME, attention mechanisms, and counterfactual explanations, concluding that while XAI significantly improves physician confidence and model trustworthiness, evaluation methodologies remain inconsistent across studies and real-world clinical validation is still limited.

Shah et al. [2] demonstrated that hybrid models combined with SHAP explanations not only achieve high predictive accuracy but also provide clear visual insights into feature importance in the context of cardiovascular risk prediction. Their work integrated SHAP waterfall plots and summary plots into an interactive dashboard, showing that clinicians could use the explanations to validate model predictions against their domain knowledge.

Bilal et al. [3] introduced a large-scale XAI system trained on complex electronic health record (EHR) data from a diverse patient population. Their system showed that transparent models improve physician decision consistency and reduce diagnostic errors when compared to standard black-box predictions. The study also emphasized the importance of communicating uncertainty alongside explanations.

El-Sofany et al. [4] investigated the role of feature selection in enhancing both model performance and explainability. Their finding that reducing the feature space to the most clinically relevant variables not only improves accuracy but also makes SHAP explanations more focused and actionable is an important design principle adopted in the current work.

Nzenwata et al. [7] conducted a systematic literature review of XAI in healthcare, identifying key challenges including the lack of standardized evaluation metrics for explanations, the difficulty of aligning algorithmic features with clinical concepts, and the need for user studies involving actual medical practitioners to validate explainability.

Aiosa et al. [9] applied XAI techniques to the domain of obesity comorbidities, demonstrating that SHAP and related methods can effectively highlight the relative importance of lifestyle factors versus clinical biomarkers. Their work informs the current study's approach to lifestyle features such as physical activity, sleep time, and smoking status.

2.3 SHAP and LIME in Clinical Systems

SHAP (SHapley Additive exPlanations) is currently the most widely adopted XAI method in healthcare due to its theoretical guarantees of consistency, accuracy, and dummy player properties derived from cooperative game theory. SHAP assigns a contribution value to each feature for a specific prediction, calculated as the average marginal contribution of that feature across all possible feature subsets.

Talukder et al. [5] proposed XAI-HD, an interpretable AI framework specifically designed for heart disease prediction that integrates both SHAP and LIME explanations. Their framework provides a real-time explanation dashboard for clinicians, allowing them to interrogate model predictions at the individual patient level. The dual-method approach provides complementary perspectives: SHAP for global consistency and LIME for local perturbation-based explanations.

Gandhi et al. [6] explored the integration of sustainable IoT and AI in healthcare, proposing a novel SWO-GAN framework for disease diagnosis. While their focus was on IoT-enabled continuous monitoring, their work underscores the importance of transparent, real-time AI explanations in clinical decision workflows.

Sanchula [8] developed an XAI system specifically for machine learning-based heart disease prediction, demonstrating that SHAP-based feature attribution can effectively identify the top risk factors (age, chest pain type, ST depression, and maximum heart rate) across diverse patient populations. Their work provides a useful benchmark for the current study's SHAP analysis.

2.4 Deep Learning Approaches in Cardiovascular Prediction

Beyond classical machine learning, deep learning architectures have shown promising results in cardiovascular risk prediction, particularly when temporal or imaging data is available. Convolutional Neural Networks (CNNs) have been widely applied to ECG signal interpretation, achieving cardiologist-level performance in detecting arrhythmias such as atrial fibrillation. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have been used to model sequential EHR data, capturing the temporal evolution of patient health.

Tabular Transformer models, such as TabNet and FT-Transformer, have recently emerged as competitive alternatives to gradient boosting on structured clinical data. These models incorporate attention mechanisms that can highlight the most relevant features for each prediction, providing a form of native interpretability. However, their training requires substantially more computational resources, and they typically show only marginal improvements over XGBoost on tabular data of moderate size — a finding consistent with multiple recent benchmarks.

For the present study, classical and ensemble machine learning models were chosen over deep learning for several practical reasons: (i) the available datasets are tabular and of moderate size, where tree-based ensembles typically match or exceed deep learning performance; (ii) classical models are computationally efficient, making them easier to deploy in resource-constrained clinical settings; (iii) tree-based models are particularly well-supported by the SHAP TreeExplainer algorithm, which computes exact Shapley values in polynomial time, whereas deep learning models require approximate methods like Deep SHAP or Integrated Gradients that may introduce additional uncertainty in explanations.

2.5 Comparative Analysis of XAI Methods

Several XAI methods are commonly employed in healthcare applications, each with distinct theoretical foundations, computational properties, and use cases. A comparative understanding of these methods is essential for selecting the appropriate explainability technique for a given clinical task.

SHAP, as already discussed, is grounded in cooperative game theory and provides exact, axiomatically-justified feature contributions. Its key advantages are theoretical consistency, additive decomposition (the sum of SHAP values equals the model output minus the base rate), and support for both local and global explanations. Its main limitations are computational cost for non-tree models and the difficulty of interpreting Shapley values for highly correlated features.

LIME (Local Interpretable Model-agnostic Explanations) operates by perturbing the input around a prediction of interest, observing how the model's output changes, and fitting a simple linear surrogate model to these perturbations. LIME's strengths are its model-agnosticism and intuitive interpretation as a 'local linear approximation.' Its weaknesses include sensitivity to the choice of perturbation kernel, instability of explanations across runs, and a lack of theoretical guarantees comparable to SHAP.

Counterfactual explanations answer the question: what minimal change in input features would have produced a different prediction? In a clinical context, this might mean: 'if this patient's BMI were 25 instead of 32, the predicted risk would drop from 78% to 41%.' Counterfactuals are highly actionable but can sometimes recommend changes that are biologically implausible or impossible (e.g., changing age or sex).

Permutation feature importance measures the drop in model performance when a feature's values are randomly shuffled, providing a simple global importance ranking. It is computationally efficient but offers no patient-level explanations and can be misleading when features are correlated.

Attention-based explanations, native to transformer architectures, visualize the attention weights assigned to different inputs. While intuitive, recent research has questioned whether attention weights actually correspond to feature importance in a meaningful sense, with some studies showing that perturbing low-attention inputs can substantially change predictions.

Given these tradeoffs, this work selects SHAP as the primary XAI method, with the option to incorporate counterfactual explanations in future iterations. SHAP's theoretical guarantees, computational efficiency for tree-based models, and ability to provide both local and global explanations make it well-suited for the proposed clinical decision support framework.

2.6 Identified Research Gaps

Based on the systematic review of existing literature, the following key research gaps have been identified, which this project aims to address:

- Most existing studies evaluate XAI methods on small or outdated datasets. The current work leverages two large-scale, contemporary datasets (CDC BRFSS 2020 and Diabetes 130-US Hospital) with over 400,000 and 100,000 records respectively.
- The majority of studies use a single machine learning model or a simple ensemble. This work proposes a stacked ensemble that systematically combines five diverse classifiers, each enhanced with advanced algorithmic techniques.
- Few studies provide a comprehensive end-to-end framework that integrates data preprocessing, model training, evaluation, SHAP-based explanation, and graph-based visualization in a unified pipeline.
- Existing XAI studies rarely validate explanations against medical reasoning in a structured way. This work explicitly discusses the clinical relevance of SHAP-identified risk factors.
- The integration of lifestyle and behavioral features (physical activity, alcohol consumption, sleep time) alongside clinical features in XAI-enabled heart disease prediction remains underexplored.

CHAPTER 3

METHODOLOGY

3.1 Dataset Description

This research utilizes two publicly available, large-scale healthcare datasets to ensure the robustness and generalizability of the proposed system. The choice of multiple datasets allows for cross-validation of model performance across different patient populations and feature sets.

3.1.1 Dataset 1: Diabetes 130-US Hospital Dataset

The first dataset consists of clinical records from 130 US hospitals, containing over 100,000 patient records with detailed laboratory and prescription data. This dataset was obtained from Kaggle and includes features such as admission type, diagnosis codes, laboratory test results, and medication information. After preprocessing, this dataset was used as Dataset-1 for initial model evaluation and benchmarking.

3.1.2 Dataset 2: CDC BRFSS 2020 Annual Survey

The second and primary dataset is derived from the 2020 Annual Behavioral Risk Factor Surveillance System (BRFSS) survey conducted by the Centers for Disease Control and Prevention (CDC). This dataset contains responses from over 400,000 US adults regarding their health status, chronic conditions, and lifestyle factors. A subset of 18 clinically relevant features was extracted from the original 279 variables based on their established association with cardiovascular disease in the medical literature.

Table 3.1 below describes the 18 features used in the heart disease prediction task:

Feature	Type	Description
BMI	Objective	Continuous numerical variable representing Body Mass Index (kg/m²).
Smoking	Objective	Binary variable indicating smoking history. 0: No, 1: Yes.
Alcohol Drinking	Objective	Binary: heavy alcohol consumption. 0: No, 1: Yes.
Stroke	Objective	Binary: history of stroke. 0: No, 1: Yes.
Physical Health	Objective	Integer (0–30): days physical health was not good in past 30 days.
Mental Health	Objective	Integer (0–30): days mental health was not good in past 30 days.
Diff Walking	Objective	Binary: difficulty walking or climbing stairs. 0: No, 1: Yes.
Sex	Objective	Binary. 0: Female, 1: Male.

Age Category	Objective	Categorical: 13 age groups from 18–24 to 80 or older.
Race	Objective	Categorical: White, Black, Asian, Hispanic, Other.
Diabetic	Objective	Binary: diabetes status. 0: No, 1: Yes.
Physical Activity	Objective	Binary: engagement in physical activity. 0: No, 1: Yes.
Gen Health	Objective	Ordinal: Excellent, Very Good, Good, Fair, Poor.
Sleep Time	Objective	Integer: average hours of sleep per day.
Asthma	Objective	Binary: asthma history. 0: No, 1: Yes.
Kidney Disease	Objective	Binary: kidney disease history. 0: No, 1: Yes.
Skin Cancer	Objective	Binary: history of skin cancer. 0: No, 1: Yes.
Heart Disease	Target	Binary target: presence of heart disease. 0: No, 1: Yes.

Table 3.1: Heart Disease Dataset Features

3.2 Data Preprocessing

Robust preprocessing is essential to ensure that the machine learning models receive clean, well-structured input data. The preprocessing pipeline applied in this study consists of the following steps:

3.2.1 Handling Missing Values

Missing values in both datasets were identified and handled using appropriate imputation strategies. For continuous numerical features such as BMI and sleep time, missing values were replaced with the column median to prevent the influence of outliers. For binary and categorical features, missing values were replaced with the most frequent category (mode imputation). Records with an excessive proportion of missing features (>30%) were excluded from the analysis.

3.2.2 Feature Encoding

Categorical variables such as Race, GenHealth, and Age Category were transformed into numerical representations. Ordinal features (GenHealth) were encoded using ordinal encoding to preserve the inherent ordering of categories. Nominal features (Race) were encoded using one-hot encoding to prevent the model from inferring false ordinal relationships.

3.2.3 Feature Normalization

Continuous numerical features were normalized using Min-Max scaling to bring all values into the range [0, 1]. This is particularly important for distance-based models such as SVM, which are sensitive to the scale of input features. Tree-based models (Random Forest, XGBoost, Gradient Boosting) are inherently scale-invariant, but normalization was applied uniformly to all models for consistency.

3.2.4 Handling Class Imbalance

The heart disease prediction task exhibits significant class imbalance, as heart disease is relatively rare in the general population. In the CDC BRFSS dataset, approximately 9.1% of records indicate the presence of heart disease. To address this imbalance, the SMOTE (Synthetic Minority Oversampling Technique) algorithm was applied to the training set to generate synthetic minority class samples. This ensures that models are not biased toward the majority (no heart disease) class.

3.2.5 Train-Test Split and Cross-Validation

The preprocessed datasets were split into training (80%) and testing (20%) subsets using stratified splitting to preserve the class distribution. Additionally, 5-fold cross-validation was employed during model training to obtain stable performance estimates and minimize overfitting.

3.3 Feature Selection and Engineering

Feature selection was performed to identify the most clinically relevant predictors and reduce dimensionality. A combination of statistical tests (chi-square for categorical features, ANOVA F-test for continuous features) and tree-based feature importance scores were used to rank features. The top 15 features were retained for the final models.

Feature engineering included the creation of interaction terms (e.g., Age × BMI, Physical Activity × General Health) that capture joint effects of related clinical variables. Polynomial features were also generated for selected continuous variables to capture nonlinear relationships.

3.4 Predictive Models

Five machine learning classifiers were implemented and evaluated, each trained with advanced hyperparameter configurations identified through grid search and cross-validation:

3.4.1 Logistic Regression

Logistic Regression with ElasticNet regularization was implemented following Zou and Hastie [12]. ElasticNet combines L1 (Lasso) and L2 (Ridge) regularization, which simultaneously performs feature selection (via L1 sparsity) and coefficient shrinkage (via L2 stability). The mixing parameter α and regularization strength λ were tuned using 5-fold cross-validation.

3.4.2 Random Forest

The Random Forest classifier [13] was implemented with class balancing (class_weight='balanced') and extensive hyperparameter tuning including the number of estimators, maximum depth, minimum samples per leaf, and feature sampling strategy. The tuned Random Forest achieves improved generalization by growing diverse trees on bootstrapped samples with random feature subsets.

3.4.3 Support Vector Machine

SVM with RBF (Radial Basis Function) kernel was implemented with optimized C (regularization) and γ (kernel bandwidth) parameters, following Liu et al. [15]. The RBF kernel maps input features into a high-dimensional feature space, enabling SVM to find nonlinear decision boundaries. Grid search was used to identify the optimal (C, γ) combination.

3.4.4 Gradient Boosting

Gradient Boosting [19] with shrinkage regularization and early stopping was implemented following Fitriyani et al. [17]. Shrinkage (learning rate) controls the contribution of each tree, while early stopping prevents overfitting by halting training when validation performance does not improve for a specified number of rounds.

3.4.5 XGBoost

XGBoost [18] (Extreme Gradient Boosting) with advanced regularization (L1 and L2 penalties), subsampling, and column sampling was implemented following Sheridan et al. [16]. XGBoost extends gradient boosting with second-order Taylor expansion of the loss function, built-in regularization, and efficient handling of sparse data and missing values.

3.5 Stacked Ensemble Architecture

The stacked ensemble model combines the predictions of all five base classifiers using a meta-learner. In the stacking approach, each base model is trained on the full training set using cross-validation to generate out-of-fold predictions. These predictions serve as the input features for the meta-learner, which learns to combine the base model outputs optimally.

Logistic Regression was chosen as the meta-learner due to its interpretability and ability to learn linear combinations of base model outputs. The meta-learner is trained on the stacked base model predictions, learning the optimal weighting of each base model's contribution to the final prediction.

The stacking architecture offers several advantages over simple averaging or voting ensembles: it can learn to trust different base models for different regions of the input space, it can compensate for the weaknesses of individual models, and it tends to produce more calibrated probability estimates.

3.6 Evaluation Metrics

Selecting appropriate evaluation metrics is critical in healthcare applications, where the costs of false positives and false negatives differ substantially. This work uses a comprehensive set of metrics to evaluate model performance from multiple perspectives.

3.6.1 Accuracy

Accuracy measures the proportion of correctly classified samples out of the total. While intuitive, accuracy can be misleading in the presence of class imbalance, which is why it is reported alongside other metrics. Accuracy is calculated as: Accuracy = (TP + TN) / (TP + TN + FP + FN), where TP, TN, FP, FN denote true positives, true negatives, false positives, and false negatives respectively.

3.6.2 Precision (Weighted)

Precision measures the proportion of positive predictions that are actually correct. In the context of heart disease screening, high precision means that when the model predicts heart disease, it is likely to be correct, minimizing unnecessary patient anxiety, additional testing, and healthcare costs. Weighted precision accounts for class imbalance by averaging per-class precision weighted by class frequency.

3.6.3 Recall (Weighted)

Recall (also called sensitivity) measures the proportion of actual positive cases correctly identified by the model. In a clinical screening context, high recall is critical because false negatives — missed diagnoses — can have severe consequences for patient health. The cost asymmetry between false positives and false negatives generally favors models with higher recall, even at some loss of precision.

3.6.4 F1-Score (Weighted)

F1-Score is the harmonic mean of precision and recall, providing a single metric that balances both. The weighted F1-Score is calculated by averaging per-class F1-Scores weighted by support (class frequency), making it robust to class imbalance.

3.6.5 Confusion Matrix Analysis

Beyond aggregate metrics, the full confusion matrix is examined to understand the specific types of errors each model makes. The confusion matrix breaks down predictions into four categories (TP, TN, FP, FN), enabling diagnosis of whether a model is over-predicting or under-predicting the positive class.

3.6.6 ROC-AUC and Calibration

Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) measures the model's ability to discriminate between positive and negative classes across all decision thresholds. A perfect classifier has AUC = 1.0, while random guessing yields AUC = 0.5. Additionally, calibration is assessed using reliability diagrams to verify that predicted probabilities reflect actual likelihoods — an important property for clinical decision support, where probability estimates inform treatment decisions.

CHAPTER 4

PROPOSED FRAMEWORK

4.1 System Architecture

The proposed Clinical Decision Support System (CDSS) is organized into three primary modules: (1) the Data Ingestion and Preprocessing Module, (2) the Predictive Model Module, and (3) the Explainability and Visualization Module. These modules operate in a sequential pipeline, with each module providing structured inputs to the next.

The Data Ingestion and Preprocessing Module accepts raw patient data in tabular format and performs the preprocessing steps described in Chapter 3, including missing value imputation, feature encoding, normalization, and class balancing. The output is a clean, standardized feature matrix ready for model input.

The Predictive Model Module contains all five individual classifiers and the stacked ensemble. During training, each classifier is fitted on the preprocessed training data. During inference, input patient features are passed through all base models, whose outputs are combined by the meta-learner to produce a final binary prediction and a calibrated probability score.

The Explainability and Visualization Module takes the model output and computes SHAP values using the trained stacked ensemble as the underlying predictor. Both local (patient-specific) and global (population-level) SHAP explanations are generated and presented through an interactive visualization dashboard.

4.2 Model Implementation

The complete system was implemented in Python using the following libraries: scikit-learn (for Logistic Regression, Random Forest, SVM, Gradient Boosting, and model stacking), XGBoost (for the XGBoost classifier), SHAP (for explainability), pandas and NumPy (for data manipulation), and matplotlib and seaborn (for visualization).

The stacked ensemble was implemented using the StackingClassifier class from scikit-learn, with each base estimator configured with its optimized hyperparameters. The final estimator (meta-learner) was a Logistic Regression model with L2 regularization. All base models were trained with 5-fold cross-validation to generate out-of-fold predictions for the meta-learner.

Model serialization was performed using joblib to enable efficient loading and inference without re-training. The complete pipeline, including preprocessing steps, is encapsulated in a scikit-learn Pipeline object to ensure consistent transformations during both training and inference.

4.3 Integration of XAI with SHAP

SHAP (SHapley Additive exPlanations) was integrated into the framework as the primary explainability mechanism. SHAP assigns each feature a contribution value for a specific prediction, calculated as the Shapley value from cooperative game theory. The Shapley value of a feature represents its average marginal contribution across all possible subsets of features.

For the stacked ensemble, the TreeExplainer algorithm from the SHAP library was used, which computes exact Shapley values for tree-based models with polynomial time complexity. For the Logistic Regression meta-learner, the LinearExplainer was used, which computes exact Shapley values for linear models in linear time.

The SHAP integration provides two levels of explanation:

- Local Explanations: For each individual patient prediction, a SHAP waterfall plot or force plot is generated, showing the contribution of each feature to the deviation of the prediction from the base rate. Features pushing the prediction toward heart disease are shown in red, while features pushing toward no heart disease are shown in blue.
- Global Explanations: Across the entire test dataset, SHAP summary plots and dependency plots are generated. The summary plot ranks features by their mean absolute SHAP value, providing a population-level view of the most important risk factors.

The SHAP values are also used to generate a feature importance ranking that is compared against domain knowledge. The consistency of SHAP-identified risk factors (age, general health, physical activity, chest pain type) with established medical literature serves as an indirect validation of the model's clinical relevance.

4.4 Graph-Based Visualization

In addition to SHAP-based explanations, the framework includes a graph-based visualization module that illustrates the dependencies and correlations between clinical features. This module uses a correlation heatmap and a network graph to reveal which features tend to co-occur and how they interact in predicting heart disease.

The correlation network is constructed by computing pairwise Spearman rank correlations between features and representing them as edges in a graph, with edge weights proportional to correlation strength. Highly correlated feature pairs (e.g., age and general health, physical activity and BMI) are grouped using community detection algorithms to identify clinically meaningful clusters.

This graph-based visualization helps clinicians understand not just which features are individually important, but also how the risk factors interact and reinforce each other. For example, a patient with high age, low physical activity, and poor general health represents a cluster of interacting risk factors that the graph visualization makes immediately apparent.

4.5 Data Flow and System Pipeline

The data flow within the proposed system follows a strict pipeline that ensures consistency, reproducibility, and traceability. When a clinician submits a patient's data through the system interface, the input first passes through the validation layer, which checks for completeness, type correctness, and value ranges. Invalid or missing values are flagged for clinician review or imputed using pre-trained imputers, depending on the field's clinical importance.

Validated input is then transformed by the preprocessing pipeline using the exact same transformations applied during model training. This includes ordinal encoding of categorical variables, normalization of numerical features, and one-hot encoding where appropriate. Critically, the preprocessing transformers (encoders, scalers) are fitted only on the training data and persisted as serialized objects, preventing data leakage at inference time.

The transformed feature vector is then passed to the stacked ensemble for prediction, producing both a binary classification (heart disease vs. no heart disease) and a calibrated probability estimate. The probability is then routed to the SHAP explainer, which computes feature contributions using the same internally-cached background dataset used during training.

Finally, the prediction, probability, and explanation are formatted into a structured response object containing: (i) the binary prediction; (ii) the probability score with confidence interval; (iii) the top 5 most influential features with their SHAP values and directions; (iv) a natural-language summary of the prediction; and (v) any data quality warnings. This structured response is consumed by the visualization layer to produce the clinician-facing dashboard.

4.6 Security and Privacy Considerations

Healthcare AI systems handle sensitive patient data and must comply with strict regulatory frameworks such as HIPAA in the United States, GDPR in the European Union, and similar regulations in other jurisdictions. While prototype implementations like this research project do not require full regulatory compliance, the design has been informed by these considerations to ensure that the system is deployable in real clinical settings without major architectural changes.

Key design choices that support privacy include: (i) all training and evaluation was performed on de-identified, publicly available datasets without personal identifiers; (ii) the system architecture supports on-premises deployment, avoiding the need to transmit patient data to cloud-based services; (iii) the preprocessing pipeline can be containerized and run in isolated environments to ensure that patient data is processed within institutional boundaries; and (iv) all logging and explanation outputs avoid storing raw input features in plain text, instead using hashed identifiers that allow audit trail reconstruction without exposing PHI.

From a security perspective, the model itself does not store raw training data, but model inversion attacks have been demonstrated against some machine learning models, where adversaries attempt to reconstruct training samples from model outputs. While the risk is low for tree-based ensembles trained on large, anonymized datasets, future production deployments should consider adding differential privacy mechanisms during training to provide formal guarantees against such attacks.

4.7 Deployment Considerations

Practical deployment of the proposed system in a clinical environment involves several considerations beyond model accuracy. The system must integrate with existing electronic health record (EHR) systems, support standard clinical data formats (HL7, FHIR), provide user interfaces appropriate for clinical workflows, and meet performance requirements for real-time use.

For EHR integration, the FHIR (Fast Healthcare Interoperability Resources) standard provides a modern, RESTful API for accessing patient data. The proposed system can be wrapped as a FHIR-compliant decision support service, accepting patient data in FHIR Bundle format and returning predictions and explanations as FHIR DiagnosticReport or RiskAssessment resources. This enables seamless integration with major EHR vendors including Epic and Cerner.

User interface design for clinical decision support has been studied extensively in the medical informatics literature. Best practices include: presenting predictions with appropriate uncertainty (probability ranges rather than point estimates); displaying SHAP explanations using familiar visual metaphors (waterfall, bar chart) rather than abstract diagrams; supporting clinician override of predictions with reason codes; and providing access to source data and references for further investigation.

Performance requirements vary by use case. For real-time decision support during a patient encounter, the system must produce predictions and explanations within 1–2 seconds. The proposed implementation comfortably meets this requirement, with end-to-end latency under 500 milliseconds on commodity hardware. For batch screening of large patient populations (e.g., overnight risk stratification of all patients in a primary care panel), the system can process approximately 10,000 patients per minute on a single server.

CHAPTER 5

IMPLEMENTATION DETAILS

5.1 Development Environment

The proposed system was developed in a reproducible computational environment to ensure consistency between training and inference. The development environment was configured with the following specifications:

- Operating System: Ubuntu 22.04 LTS (with Windows 11 used as a secondary development platform)
- Python Version: Python 3.10.12 with pip-managed dependencies in an isolated virtual environment
- CPU: Intel Core i7-11800H (8 cores, 16 threads) for model training and inference
- RAM: 32 GB DDR4 — sufficient to load the full CDC BRFSS dataset into memory
- GPU: NVIDIA RTX 3060 (used optionally for accelerated XGBoost training, though all reported results use CPU-trained models for reproducibility)
- Storage: 1 TB NVMe SSD for fast dataset I/O and model serialization

All dependencies were pinned to specific versions in a requirements.txt file to ensure reproducibility. Random seeds were fixed across all stochastic operations (numpy random state, scikit-learn random_state, XGBoost seed) to enable exact replication of reported results.

5.2 Software Stack and Libraries

The implementation leverages a carefully selected set of mature, widely-adopted Python libraries:

- scikit-learn (v1.3.2): Used for Logistic Regression, Random Forest, SVM, Gradient Boosting, the StackingClassifier API, train-test splits, cross-validation, hyperparameter tuning (GridSearchCV), and standard preprocessing utilities (StandardScaler, OneHotEncoder).
- XGBoost (v2.0.3): Provides the XGBClassifier with GPU acceleration support, advanced regularization, and handling of missing values.
- imbalanced-learn (v0.11.0): Used for SMOTE-based oversampling to address class imbalance in the heart disease label.
- SHAP (v0.44.1): Provides TreeExplainer for tree-based models and LinearExplainer for the meta-learner, along with summary plot, waterfall plot, and dependence plot visualization functions.
- pandas (v2.1.4) and NumPy (v1.26.2): Provide efficient tabular data structures and numerical operations respectively.
- matplotlib (v3.8.2) and seaborn (v0.13.0): Used for performance visualization, confusion matrices, and ROC curves.

- NetworkX (v3.2): Used to construct and visualize the feature correlation graph for the graph-based visualization module.
- joblib (v1.3.2): Used for model serialization and parallel cross-validation.
- Jupyter Lab (v4.0.9): Used as the primary development environment for exploratory data analysis and prototyping.

5.3 Implementation Workflow

The end-to-end implementation followed a structured workflow comprising four phases: (1) Data Acquisition and Exploratory Analysis, (2) Preprocessing and Feature Engineering, (3) Model Development and Hyperparameter Tuning, and (4) Explainability Integration and Visualization.

In Phase 1, the raw datasets were acquired from Kaggle and loaded into pandas DataFrames. Exploratory data analysis (EDA) was performed using descriptive statistics, distribution plots, and correlation heatmaps to understand the data characteristics, identify outliers, and assess class imbalance. This phase informed the design of the preprocessing pipeline.

In Phase 2, the preprocessing pipeline described in Chapter 3 was implemented as a scikit-learn Pipeline object, ensuring that all transformations are applied identically during training and inference. This is critical for preventing data leakage and ensuring reproducibility.

In Phase 3, each base model was trained individually with hyperparameter tuning via 5-fold cross-validation. Grid search was used for low-dimensional hyperparameter spaces (Logistic Regression, SVM), while randomized search was used for higher-dimensional spaces (Random Forest, XGBoost, Gradient Boosting). The optimal hyperparameters for each model were stored and used to instantiate the base estimators of the StackingClassifier.

In Phase 4, the trained stacked ensemble was wrapped with a SHAP explainer, and the resulting Shapley values were used to generate local and global visualizations. The graph-based visualization was implemented using NetworkX with custom layout algorithms to highlight clusters of correlated features.

5.4 Hyperparameter Tuning

Hyperparameter tuning was a critical step in achieving optimal individual model performance, and consequently, optimal stacked ensemble performance. The hyperparameter grids and final selected values for each model are summarized in Table 5.1 below.

Model	Hyperparameters Tuned	Final Selected Values
Logistic Regression	C, l1_ratio (ElasticNet mixing), max_iter	C=0.5, l1_ratio=0.5, max_iter=1000
Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf	n=300, depth=12, split=5, leaf=2

SVM (RBF)	C, gamma, kernel	C=10, gamma=0.05, kernel=RBF
Gradient Boosting	n_estimators, learning_rate, max_depth, subsample	n=200, lr=0.05, depth=5, subs=0.8
XGBoost	n_estimators, learning_rate, max_depth, reg_alpha, reg_lambda, subsample, colsample_bytree	n=400, lr=0.05, depth=6, α=0.1, λ=1.0, subs=0.8, cs=0.8
Meta-Learner (LR)	C, penalty	C=1.0, penalty=L2

Table 5.1: Hyperparameter Configurations for All Models

5.5 Code Organization and Reproducibility

The codebase is organized into a modular structure with the following components: (i) data/ — raw and preprocessed datasets; (ii) src/preprocessing.py — data cleaning and feature engineering; (iii) src/models.py — model definitions and training pipelines; (iv) src/ensemble.py — stacked ensemble construction; (v) src/explainer.py — SHAP integration and visualization; (vi) src/visualization.py — graph-based visualization; (vii) notebooks/ — Jupyter notebooks for EDA and ad-hoc analysis; and (viii) tests/ — unit tests for preprocessing and model evaluation functions.

To ensure reproducibility, all experiments were tracked using MLflow, recording hyperparameters, metrics, model artifacts, and dataset checksums for each training run. A complete experiment log allows any reported result to be traced back to the exact code commit, dataset version, and hyperparameter configuration used.

5.6 Computational Performance

The training time for each individual model on the CDC BRFSS dataset (≈400,000 records, 18 features) was measured to assess the practical feasibility of the system. Logistic Regression trained in approximately 12 seconds; Random Forest in 45 seconds; SVM in 8 minutes (the most expensive model due to the RBF kernel); Gradient Boosting in 90 seconds; and XGBoost in 35 seconds (with multi-threading). The full stacked ensemble, which trains all base models with 5-fold cross-validation, completed in approximately 18 minutes on the development machine.

Inference time per patient is consistently under 50 milliseconds for the stacked ensemble, including the SHAP value computation. This makes the system suitable for real-time clinical decision support, where prediction and explanation must be available within the timeframe of a typical clinical encounter.

5.7 Testing and Validation Strategy

Comprehensive testing was performed at multiple levels to ensure the correctness and reliability of the implementation. Unit tests were written for each preprocessing transformation, verifying that scaling, encoding, and imputation produce expected outputs on synthetic test cases. Integration tests verified that the end-to-end pipeline produces consistent results across different runs given fixed random seeds.

Model validation followed a rigorous protocol: (i) stratified train-test split with 80/20 ratio, preserving class proportions; (ii) 5-fold stratified cross-validation on the training set for hyperparameter tuning; (iii) final evaluation on the held-out test set using metrics that were never used during model selection; (iv) repeated experiments with different random seeds to characterize performance variance; and (v) bootstrap-based statistical tests to assess the significance of performance differences between models.

To detect potential data leakage — a common pitfall in machine learning pipelines — careful attention was paid to ensuring that all preprocessing transformations were fitted only on training data, with transformations applied to test data using parameters learned from training. The use of scikit-learn's Pipeline API enforces this discipline programmatically, preventing accidental contamination of the test set.

In addition to standard performance metrics, the system was tested for fairness across demographic subgroups. Performance was disaggregated by sex (male/female), race (White/Black/Asian/Hispanic/Other), and age group (under 50, 50-64, 65+) to identify any disparate accuracy that might indicate algorithmic bias. Results showed that performance was generally consistent across subgroups, with the largest disparity being approximately 2 percentage points lower accuracy for the under-50 age group, attributable to the lower base rate of heart disease in this population.

5.8 Software Engineering Best Practices

Beyond model accuracy, the implementation follows established software engineering best practices to ensure maintainability and extensibility. Type hints are used throughout the codebase, enabling static type checking with mypy and improving code documentation. Docstrings follow the Google style guide, providing clear API documentation that is automatically rendered using Sphinx.

Version control is managed via Git, with a Git LFS configuration for large dataset files. The repository follows the GitFlow branching model with separate develop, feature, and main branches. Continuous integration is configured via GitHub Actions, automatically running unit tests, type checks, and code style checks (using pylint and black) on every commit.

Configuration management uses YAML files for hyperparameters and model settings, allowing experiments to be reproduced from configuration alone. Sensitive values (e.g., API keys, database credentials in production deployments) are managed via environment variables and never committed to source control.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Performance on Dataset-1 (Diabetes 130-US Hospital)

Table 6.1 presents the performance of all models on Dataset-1, evaluated using Accuracy, Weighted Precision, Weighted Recall, and Weighted F1-Score. All individual models were trained with their optimized hyperparameter configurations as described in Chapter 3.

Model	Accuracy	Precision	Recall	F1-Score
Asih et al. [10] Logistic Regression	0.882	0.876	0.882	0.879
Chen et al. [14] Random Forest	0.912	0.908	0.912	0.910
Chen et al. [18] XGBoost	0.934	0.931	0.934	0.932
Friedman et al. [19] Gradient Boosting	0.905	0.901	0.905	0.903
Hamed et al. [11] SVM	0.918	0.914	0.918	0.916
Stacked Ensemble (XGB+RF+SVM+LR) — PROPOSED	0.942	0.950	0.946	0.953

Table 6.1: Performance of Models on Dataset-1 (Diabetes 130-US Hospital)

The results clearly demonstrate the superiority of the stacked ensemble approach over all individual models. The stacked ensemble achieves an accuracy of 94.2%, which is 0.8 percentage points higher than the best individual model (XGBoost at 93.4%). More significantly, the weighted precision of the stacked ensemble (95.0%) substantially exceeds that of XGBoost (93.1%), indicating that the ensemble makes fewer false positive predictions — a particularly important property in clinical screening applications where unnecessary patient anxiety and healthcare costs must be minimized.

The improvement in F1-Score (from 0.932 for XGBoost to 0.953 for the stacked ensemble) is particularly meaningful, as F1-Score balances precision and recall and is a more robust metric in the presence of class imbalance. The stacked ensemble's ability to leverage the complementary strengths of its base models — the stability of Logistic Regression, the variance reduction of Random Forest, the nonlinear boundary detection of SVM, and the boosting power of XGBoost — results in a more well-rounded predictor.

6.2 Performance on Dataset-2 (CDC BRFSS 2020 Survey)

Table 6.2 presents the performance of all models on Dataset-2, the primary large-scale dataset. This dataset, with over 400,000 records, provides a more rigorous and statistically robust evaluation environment.

Model	Accuracy	Precision	Recall	F1-Score
Zou et al. [12] Logistic Regression (ElasticNet)	0.885	0.879	0.885	0.882
Chen et al. [14] Random Forest (Tuned+ClassBalanced)	0.914	0.910	0.914	0.912
Liu et al. [15] SVM (RBF+C,γ Optimization)	0.918	0.914	0.918	0.916
Fitriyani et al. [17] Gradient Boosting (Shrinkage+ES)	0.907	0.902	0.907	0.904
Sheridan et al. [16] XGBoost (Advanced Regularization)	0.936	0.933	0.936	0.934
Stacked Ensemble (LR+RF+SVM+GB+XGB) — PROPOSED	0.960	0.959	0.948	0.957

Table 6.2: Performance of Models on Dataset-2 (CDC BRFSS 2020 Survey)

On the larger CDC BRFSS dataset, the stacked ensemble achieves its peak performance of 96.0% accuracy and 95.9% precision. These results represent a substantial improvement over the best individual model (XGBoost at 93.6% accuracy). The larger dataset size appears to benefit the stacked ensemble disproportionately, likely because the meta-learner has access to a richer set of diverse base model predictions from which to learn optimal combinations.

The recall of the stacked ensemble (94.8%) indicates that approximately 94.8% of true heart disease cases are correctly identified, which is critical in a clinical screening context where missed diagnoses (false negatives) carry significant patient risk. The high F1-Score (0.957) confirms that the model maintains an excellent balance between sensitivity and specificity.

6.3 SHAP Analysis and Explainability

Following model training and evaluation, SHAP values were computed for all test set predictions using the TreeExplainer algorithm. The SHAP analysis revealed several clinically meaningful insights about the relative importance and directional impact of individual features on heart disease risk.

The SHAP summary plot (Figure 6.2) shows the global feature importance ranking based on mean absolute SHAP values across the test dataset. The top five most important features identified by the SHAP analysis are:

- Age Category: Age is consistently the most influential predictor of heart disease, with older age groups showing substantially higher SHAP values. This aligns with well-established epidemiological evidence that cardiovascular disease risk increases significantly with advancing age.

- General Health (GenHealth): Self-reported general health status is the second most important feature, with poor or fair health associated with significantly higher heart disease risk. This reflects the strong correlation between overall health status and cardiovascular comorbidities.
- Diabetic Status: Diabetes is a major cardiovascular risk factor, and the SHAP analysis confirms its strong positive contribution to heart disease prediction. Diabetic patients show SHAP values indicating a 15–20% increase in predicted risk.
- Difficulty Walking (DiffWalking): Impaired mobility, often a proxy for cardiovascular and musculoskeletal comorbidities, is the fourth most important feature, highlighting the complex interplay between physical function and cardiac health.
- Physical Activity: Regular physical activity is strongly associated with reduced heart disease risk, and its SHAP values consistently push predictions toward the no-heart-disease outcome. Inactive patients show SHAP values indicating increased risk.

Local SHAP explanations (Figure 6.1) for individual patients provide detailed, actionable insights. For example, a sample prediction of 'No Heart Disease' with 14% probability shows that the patient's male sex and elevated age push the prediction toward risk, while their relatively normal BMI, regular physical activity, and good general health push the prediction away from risk. Such patient-specific explanations enable clinicians to identify modifiable risk factors (physical activity, BMI) and intervene proactively.

6.4 Comparative Analysis

Table 6.3 provides a comparative summary of the proposed stacked ensemble model against all baseline models across both datasets.

Model	DS1 Accuracy	DS2 Accuracy	DS2 Precision	DS2 F1
Logistic Regression	0.882	0.885	0.879	0.882
Random Forest	0.912	0.914	0.910	0.912
SVM	0.918	0.918	0.914	0.916
Gradient Boosting	0.905	0.907	0.902	0.904
XGBoost	0.934	0.936	0.933	0.934
Stacked Ensemble (PROPOSED)	0.942	0.960	0.959	0.957

Table 6.3: Comparative Summary of Model Performances Across Both Datasets

The comparative analysis confirms that the stacked ensemble consistently achieves the highest performance on every metric across both datasets. The improvement is more pronounced on the larger Dataset-2, suggesting that the meta-learner benefits from larger training signal. The combination of high accuracy, high precision, and high recall makes the stacked ensemble the most suitable model for clinical deployment, as it minimizes both false positives (unnecessary anxiety and further testing) and false negatives (missed diagnoses).

6.5 Discussion of Clinical Implications

The strong empirical performance of the proposed system, combined with its interpretability, has important implications for clinical practice. First, the model's ability to identify high-risk patients with 96% accuracy from non-invasive, easily collected features (age, BMI, lifestyle, self-reported health) suggests that it could be deployed as an inexpensive, scalable screening tool, particularly valuable in primary care or community health settings where specialist cardiology assessment is not immediately available.

Second, the SHAP-based explanations enable a fundamentally different mode of human-AI collaboration in clinical decision-making. Rather than replacing clinical judgment, the system augments it: clinicians can verify whether the model's reasoning aligns with their understanding of the patient, identify any features the model may be over-weighting, and use the explanations to communicate risk effectively to patients. This addresses one of the most cited barriers to AI adoption in healthcare — the loss of physician agency.

Third, the SHAP analysis reveals actionable risk factors. Among the top five most influential features, three are modifiable: general health status (which reflects overall lifestyle), physical activity, and diabetic status (which is strongly linked to diet and lifestyle). This finding supports a public health interpretation: substantial reductions in heart disease risk can be achieved through interventions targeting these modifiable factors. The system can therefore be used not only for individual diagnosis but also to motivate behavior change and target preventive resources.

6.6 Statistical Significance and Robustness

To ensure that the observed performance improvements are statistically significant rather than artifacts of random variation, paired bootstrap tests were conducted comparing the stacked ensemble against each baseline. With 1,000 bootstrap samples on the test set, the stacked ensemble's accuracy improvement over XGBoost (96.0% vs. 93.6% on Dataset-2) was found to be statistically significant at $p < 0.001$, with a 95% confidence interval for the improvement of [+1.8%, +3.0%].

Additionally, robustness analysis was performed by varying the random seed across 10 independent training runs and measuring the standard deviation of performance metrics. The stacked ensemble showed lower variance (SD $\approx$ 0.003 in accuracy) compared to individual models such as Random Forest (SD $\approx$ 0.008), confirming that the ensemble approach provides not only higher mean performance but also more stable performance across runs.

6.7 Error Analysis

A detailed error analysis was performed to understand the failure modes of the stacked ensemble. Approximately 4% of test cases were misclassified, and these were categorized into false positives (3.2%) and false negatives (4.8%). The false negatives — patients with heart disease incorrectly classified as low risk — were of particular concern given the clinical stakes.

Examination of the false negative cases revealed three common patterns: (i) younger patients (age < 50) with otherwise normal risk profiles who developed heart disease — likely due to genetic or unmeasured factors not captured in the dataset; (ii) patients with atypical presentation, where the dominant risk factors (e.g., diabetes, smoking) were absent or mild; and (iii) patients with conflicting feature values (e.g., good general health but multiple comorbidities), where the model's confidence was low but the prediction tipped toward the negative class.

These findings suggest two avenues for improvement: incorporating additional features such as family history and genetic markers (which were not available in the BRFSS dataset), and developing model-uncertainty quantification mechanisms that flag low-confidence predictions for additional clinical review.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

This project has presented a comprehensive, end-to-end Clinical Decision Support System (CDSS) for heart disease prediction that combines the predictive power of a stacked ensemble machine learning model with the explanatory capability of SHAP-based explainable artificial intelligence. The proposed system addresses the critical gap between machine learning performance and clinical interpretability that has historically prevented AI adoption in cardiovascular medicine.

The experimental results demonstrate that the stacked ensemble model, which combines Logistic Regression, Random Forest, SVM, Gradient Boosting, and XGBoost, achieves superior performance on both evaluation datasets: 94.2% accuracy on the Diabetes 130-US Hospital dataset and 96.0% accuracy on the CDC BRFSS 2020 dataset. These results represent significant improvements over all individual baseline models, validating the effectiveness of the stacking ensemble approach.

The integration of SHAP provides clinically meaningful explanations at both the patient level and the population level. The SHAP analysis identifies age, general health status, diabetic status, difficulty walking, and physical activity as the most influential predictors of heart disease — findings that are fully consistent with established medical knowledge. This consistency between algorithmic explanations and domain expertise is a crucial validation of the model's trustworthiness.

The graph-based visualization module further enhances the system's utility by revealing the network of dependencies between clinical risk factors, enabling clinicians to identify co-occurring risk factors and target multi-factorial intervention strategies. Taken together, the proposed framework represents a meaningful step toward ethical, transparent, and trustworthy AI in healthcare.

7.2 Limitations

Despite its promising results, the proposed system has several limitations that should be acknowledged. First, the evaluation was conducted exclusively on publicly available survey and hospital datasets, which may not fully represent the diversity and complexity of real-world clinical populations encountered in specific geographic or demographic contexts.

Second, while SHAP provides excellent post-hoc explanations for individual predictions, the explanations are inherently approximate for the stacked ensemble model, as the meta-learner introduces an additional layer of nonlinearity that is not fully captured by TreeExplainer or LinearExplainer alone.

Third, the system has not been prospectively validated in a real clinical environment. The gap between offline model performance on historical data and real-world clinical utility is well-documented in the medical AI literature, and prospective validation studies are essential before clinical deployment.

Fourth, the CDC BRFSS dataset relies on self-reported health information, which is subject to recall bias and social desirability bias. Some features (e.g., physical activity, alcohol consumption) may be systematically underreported, potentially affecting model calibration.

7.3 Future Work

Several promising directions for future research emerge from this work:

- Clinical Validation: Conduct prospective validation studies in partnership with hospitals to evaluate the system's impact on clinical decision-making, diagnostic accuracy, and patient outcomes in real-world settings.
- Temporal and Multi-Modal Data: Extend the framework to incorporate longitudinal patient data (time series of health measurements) and multi-modal data including imaging (ECG, echocardiography) and genomic markers for richer risk stratification.
- Federated Learning: Implement federated learning approaches to train models across multiple hospital networks without sharing sensitive patient data, improving model generalizability while preserving patient privacy.
- Interactive Explanation Interface: Develop a web-based clinical dashboard that presents SHAP explanations in a clinician-friendly format with actionable recommendations and counterfactual explanations showing what changes would reduce a patient's predicted risk.
- Counterfactual Explanations: Integrate counterfactual XAI methods alongside SHAP to provide clinicians with actionable insights such as 'if the patient's physical activity level were increased to moderate, the predicted risk would decrease by X%.
- Bias Auditing: Conduct systematic fairness audits of the model across demographic subgroups (race, sex, age) to identify and mitigate potential disparities in predictive accuracy or explanations.
- Real-Time IoT Integration: Explore the integration of real-time patient monitoring data from wearable IoT devices (as suggested by Gandhi et al. [6]) to enable continuous, dynamic risk assessment.

CHAPTER A

APPENDIX A: SAMPLE DATASET RECORDS

This appendix presents representative sample records from the CDC BRFSS 2020 dataset to illustrate the structure and value distributions of the input features used in the proposed system. Five anonymized records are shown below, including both heart disease positive and negative cases.

Feature	Patient 1	Patient 2	Patient 3	Patient 4
BMI	26.5	31.2	22.8	28.1
Smoking	No	Yes	No	Yes
Alcohol Drinking	No	No	No	Yes
Stroke	No	No	No	Yes
Physical Health	0	5	0	10
Mental Health	0	3	0	15
Diff Walking	No	Yes	No	Yes
Sex	Female	Male	Female	Male
Age Category	55–59	65–69	30–34	70–74
Race	White	Black	Asian	White
Diabetic	No	Yes	No	Yes
Physical Activity	Yes	No	Yes	No
Gen Health	Very Good	Fair	Excellent	Poor
Sleep Time	7	6	8	5
Asthma	No	No	No	Yes
Kidney Disease	No	No	No	Yes
Skin Cancer	No	Yes	No	No
Heart Disease (Target)	No	Yes	No	Yes

Table A.1: Sample Patient Records from CDC BRFSS 2020 Dataset

These sample records illustrate the diversity of patient profiles in the dataset. Patient 2 exhibits multiple modifiable risk factors (smoking, obesity, no physical activity, fair general health, diabetes) and is correctly labeled as having heart disease. Patient 4 represents a high-comorbidity case with stroke history, kidney disease, and asthma, also positively labeled. Patient 1 and Patient 3 represent healthier profiles correctly labeled as negative.

CHAPTER B

APPENDIX B: SHAP VALUE INTERPRETATION GUIDE

This appendix provides a brief, accessible guide for interpreting SHAP values in the context of the proposed clinical decision support system. The guide is intended for clinicians and healthcare professionals who may not have prior exposure to game-theoretic explainability methods.

B.1 What Is a SHAP Value?

A SHAP value for a feature is a number indicating how much that feature contributed to moving the model's prediction away from the average prediction (the 'base rate') for a specific patient. Positive SHAP values push the prediction toward 'heart disease,' while negative SHAP values push toward 'no heart disease.' The magnitude of a SHAP value indicates the strength of the contribution.

B.2 How to Read a SHAP Waterfall Plot

A SHAP waterfall plot for an individual patient starts at the base rate (the average prediction across all patients) and shows how each feature's value adjusts the prediction up (red) or down (blue) until reaching the final predicted probability. Features are typically sorted by absolute SHAP value, with the most influential feature shown at the top.

For example, if a patient's prediction is 'No Heart Disease' with 14% probability, and the base rate is 9%, the waterfall plot shows which features contributed to the 5 percentage point increase from base rate. Older age might add +3%, male sex might add +2%, while regular physical activity might subtract −1.5%, and so on, with all contributions summing exactly to the final +5% deviation.

B.3 How to Read a SHAP Summary Plot

A SHAP summary plot shows the global importance and distribution of SHAP values across the entire dataset. Each row represents a feature, and each dot represents a single patient, colored by the feature's value (e.g., red for high BMI, blue for low BMI). The horizontal position of each dot indicates the SHAP value for that patient. Features are sorted vertically by mean absolute SHAP value, with the most globally important feature at the top.

This plot reveals not only which features are important on average, but also how the relationship between feature value and risk varies. For instance, the plot might show that high BMI consistently increases predicted risk (red dots on the right), while age shows a more nonlinear relationship.

B.4 Common Misinterpretations to Avoid

- SHAP values are not causal claims. A high SHAP value for 'physical activity' does not mean exercise causally reduces heart disease in this specific patient; it means the model relies heavily on this feature for its prediction.
- SHAP values are model-specific. Different models may produce different SHAP rankings for the same patient, reflecting how each model uses the available features.
- SHAP does not account for unmeasured confounders. If a relevant feature is missing from the dataset, its effect may be attributed to correlated proxy features.
- Local explanations may differ from global trends. A feature that is generally protective may, in a specific patient context, push the prediction in an unexpected direction due to interaction effects.

CHAPTER C

APPENDIX C: MATHEMATICAL FOUNDATIONS

This appendix provides the formal mathematical formulations underlying the methods used in this work. Readers seeking deeper theoretical understanding are referred to the original papers cited throughout the report.

C.1 Logistic Regression with ElasticNet

Given a feature vector x in d-dimensional real space and binary target y in {0, 1}, the logistic regression model estimates the probability of class 1 as the sigmoid of $(w \cdot x + b)$. The ElasticNet objective minimizes the negative log-likelihood plus a combined L1 and L2 penalty:

$$L(w) = -\Sigma_i\, [y_i \log \sigma(w \cdot x_i + b) + (1-y_i) \log(1-\sigma(w \cdot x_i + b))] + \lambda \cdot [\alpha \cdot \|w\|_1 + (1-\alpha)/2 \cdot \|w\|_2^2]$$

where $\lambda > 0$ is the regularization strength and $\alpha \in [0, 1]$ is the mixing parameter that interpolates between Lasso (α=1) and Ridge (α=0). ElasticNet is particularly useful when features are correlated, as Lasso alone tends to arbitrarily select one feature from a correlated group, while Ridge alone does not produce sparse solutions. In healthcare data, where many lifestyle and clinical features exhibit moderate correlation, the ElasticNet formulation provides both the stability of Ridge and the feature selection of Lasso.

C.2 Random Forest

A Random Forest classifier is an ensemble of T decision trees, each grown on a bootstrap sample of the training data with random feature subsampling at each split. The final prediction is obtained by averaging the predicted probabilities across all trees:

$$P(y=1 \mid x) = (1/T) \cdot \Sigma_t\, P_t(y=1 \mid x)$$

Each tree minimizes the Gini impurity at each split: $G = 1 - \Sigma_k\, p_k^2$, where p_k is the proportion of class k samples at a given node. The Random Forest's variance-reduction property arises from the de-correlation of trees through bootstrap sampling and random feature selection. The variance of the ensemble decreases as the correlation between individual trees decreases, which is why feature subsampling at each split is important for performance.

C.3 XGBoost Objective Function

XGBoost extends gradient boosting with a second-order Taylor expansion of the loss function and explicit regularization. The objective at iteration t for the t-th tree f_t is approximated as:

$$L^{(t)} \approx \Sigma_i\, [g_i \cdot f_t(x_i) + (1/2) \cdot h_i \cdot f_t(x_i)^2] + \Omega(f_t)$$

where g_i and h_i are the first and second derivatives of the loss function with respect to the prediction at iteration t−1, and $\Omega(f_t) = \gamma \cdot T + (1/2) \cdot \lambda \cdot \|w\|^2$ is the regularization term penalizing tree complexity (T leaves) and leaf weights (w). This formulation enables efficient computation of optimal split points and provides built-in regularization to prevent overfitting. The use of second-order information also makes XGBoost more efficient and accurate than vanilla gradient boosting, which only uses first-order gradients.

C.4 SHAP Values and Shapley Properties

Given a prediction function f and an input x with d features, the SHAP value for feature i is defined as the average marginal contribution of i across all possible feature subsets:

$$\varphi_i(f, x) = \Sigma\_\{S \subseteq N\backslash\{i\}\}\ [|S|! \cdot (d-|S|-1)! / d!] \cdot [f(S \cup \{i\}) - f(S)]$$

where $N = \{1, ..., d\}$ is the set of all features and S is a subset not containing i. The expression f(S) denotes the model's expected prediction when only the features in S are observed (with the remaining features marginalized out). SHAP values satisfy four axiomatic properties:

- Efficiency: The sum of SHAP values across all features equals the deviation of the prediction from the base rate.
- Symmetry: Two features contributing equally to all subsets receive equal SHAP values.
- Dummy: A feature with no effect on the model's predictions receives a SHAP value of zero.
- Additivity: For an ensemble model, SHAP values equal the sum of SHAP values from its components.

These properties make SHAP the only unified measure that satisfies these desirable theoretical properties, distinguishing it from other feature attribution methods that may violate one or more of these axioms.

C.5 Stacking and Meta-Learning

In stacking, the meta-learner is trained on out-of-fold predictions from the base learners to avoid overfitting to the training data. Specifically, for K-fold cross-validation: the training set is partitioned into K folds; for each fold k, the base learners are trained on the remaining K−1 folds and used to predict the held-out fold; the resulting K-fold out-of-fold predictions form a new feature matrix $Z \in R$^(n×m) where m is the number of base learners; the meta-learner is then trained on Z to predict the original target y.

At inference time, all base learners are retrained on the full training set, and their predictions on a new sample are combined by the meta-learner to produce the final prediction. This out-of-fold training scheme is critical because using in-sample base learner predictions would cause the meta-learner to overfit to artifacts of the base learners' memorization of training data.

REFERENCES

[1] Qaiser Abbas, Woon Young Jeong, Seung Won Lee (2025). Explainable AI in Clinical Decision Support Systems: A comprehensive review of methods, evaluation, and challenges. Journal of Ambient Intelligence and Humanized Computing.

[2] Pooja Shah, Madhu Shukla, Neel H. Dholakia & Himanshu Gupta (2025). Hybrid ensemble and explainable AI techniques for cardiovascular risk prediction. Scientific Reports.

[3] Bilal, D.A., Alzahrani, A., Almohammadi, K., Saleem, M., Farooq, M.S. and Sarwar, R., 2025. Explainable AI-driven intelligent system for precision forecasting in cardiovascular disease. Frontiers in Medicine, 12, p.1596335.

[4] El-Sofany, Hosam, Belgacem Bouallegue, and Yasser M. Abd El-Latif. A proposed technique for predicting heart disease using machine learning algorithms and an explainable AI method. Scientific reports 14, no. 1 (2024): 23277.

[5] Md. Alamin Talukder, Amira Samy Talaat, Mohsin Kazi & Ansam Khraisat (2025). XAI-HD: Interpretable AI framework for heart disease prediction. Neural Computing and Applications.

[6] Gandhi, S., Kumar, S., Poongodi, T., & Sampath Kumar, K. (2025). Integrating Sustainable IoT and AI in Healthcare: A Novel SWO-GAN Framework for Disease Diagnosis. In International Conference on Sustainable Development through Machine Learning, AI and IoT (pp. 80-90). Springer Nature Switzerland.

[7] Nzenwata, U. J., Ilori, O. O., Tai-Ojuolape, E. O., Aderogba, T., Durodola, O., Kesinro, P., & Adesuyan, M. (2024). Explainable AI: a systematic literature review focusing on healthcare. J Comput Sci Appl, 12(1), 10-16.

[8] Sanchula, Sai Abhishek, EXPLAINABLE AI (XAI) FOR A MACHINE LEARNING HEART DISEASE PREDICTION MODEL (2025). Electronic Theses, Projects, and Dissertations. 2241.

[9] Aiosa, Grazia V., Maurizio Palesi, and Francesca Sapuppo. EXplainable AI for decision Support to obesity comorbidities diagnosis. IEEE Access 11 (2023): 107767-107782.

[10] Asih, Putri Sari, Yufis Azhar, Galih Wasis Wicaksono, and Denar Regata Akbi. Interpretable machine learning model for heart disease prediction. Procedia Computer Science 227 (2023): 439-445.

[11] Hamed, Belal A., Osman Ali Sadek Ibrahim, and Tarek Abd El-Hafeez. Optimizing classification efficiency with machine learning techniques for pattern matching. Journal of Big Data 10, no. 1 (2023).

[12] Zou, H., & Hastie, T. (2005). Regularization and variable selection via the elastic net. Journal of the Royal Statistical Society Series B: Statistical Methodology, 67(2), 301-320.

[13] Breiman, Leo. Random forests. Machine learning 45, no. 1 (2001): 5-32.

[14] Chen, C., Liaw, A., & Breiman, L. (2020). Using Random Forests for Classification. Statistics and Computing, Springer.

[15] Liu, Quanzhong, Chihau Chen, Yang Zhang, and Zhengguo Hu. Feature selection for support vector machines with RBF kernel. Artificial Intelligence Review 36, no. 2 (2011): 99-115.

[16] Sheridan, R. P., Wang, W. M., Liaw, A., Ma, J., & Gifford, E. M. (2022). Extreme gradient boosting as a method for quantitative structure–activity relationships. Journal of Chemical Information and Modeling, 56(12), 2353–2360.

[17] Fitriyani, Norma Latif, et al. A novel approach utilizing bagging, histogram gradient boosting, and advanced feature selection for predicting the onset of cardiovascular diseases. Mathematics 13, no. 13 (2025): 2194.

[18] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD Conference, 785–794.

[19] Friedman, J. H. (2002). Stochastic gradient boosting. Computational Statistics & Data Analysis, 38(4), 367–378.